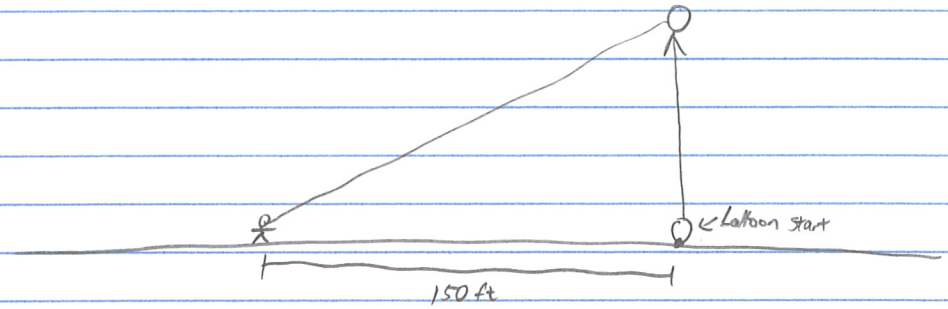
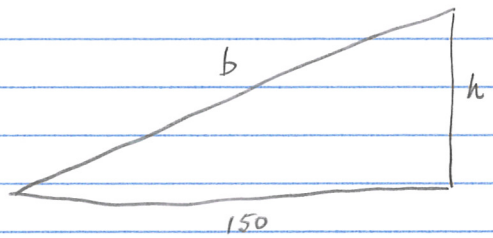
(1)





$h(t)$ height of balloon } at time t
$b(t)$ "between" distance }

$$b^2 = 150^2 + h^2$$
$$b^2 = 22500 + h^2$$
$$2b \cdot \frac{db}{dt} = 2h \cdot \frac{dh}{dt}$$

At time t, ... $\frac{dh}{dt} = 8$ $h = 50$

$$b^2 = 150^2 + 50^2$$
$$b^2 = 22500 + 2500$$
$$b = 50\sqrt{10}$$

$$2(50\sqrt{10}) \cdot \frac{db}{dt} = 2(50) \cdot 8$$

$$\frac{db}{dt} = \frac{2 \cdot 50 \cdot 8}{2 \cdot 50\sqrt{10}} = \frac{8}{\sqrt{10}} \approx 2.53$$

At the instant when $h = 50$, the distance btwn you and the observer is increasing at 2.53 ft/sec.